EX.NO: 08
DATE: 28/9/24

NMAP LIVE HOST DISCOVERY
ARP, ICMP and TCP/UDP PING SCANS

AIM:

To discovers lives thats on a network using Nmap that discovery scans - and to understand which protocol operate at which layer of the TCP/IP model, and when each on applies

ARP Host Discovery:

```
$ sudo nmap -PR -Sn 10.200.6.0/24
```

o/p:-

```
starting Nmap 7.80 ( https://nmap.org )
Nmap Scan repost for 10200.6.)
Host is up.
Nmap done: 256 IP address (1 host up) Scanned in
           10.51 seconds
```

ICMP Host Discovery:

T.11 ICMP Echo (-PE):

```
$ sudo nmap -PE -Sn 10.200.6.0/24
Nmap Scan repost for 10.200.6.1
Host is up (0.69s latency)
Nmap Scan repost for 10.200.6.50
Host is up (0.070s latency)
```

ICMP Timestamp (-PP):

```
$ nmap -PP -Sn 10.200.6.0/24
Nmap Scan repost for 10.200.6.50
Host is up (0.06as latency)
Nmap Scan report for 10.200.6.250
Host is up (0.0698 Latency)
```

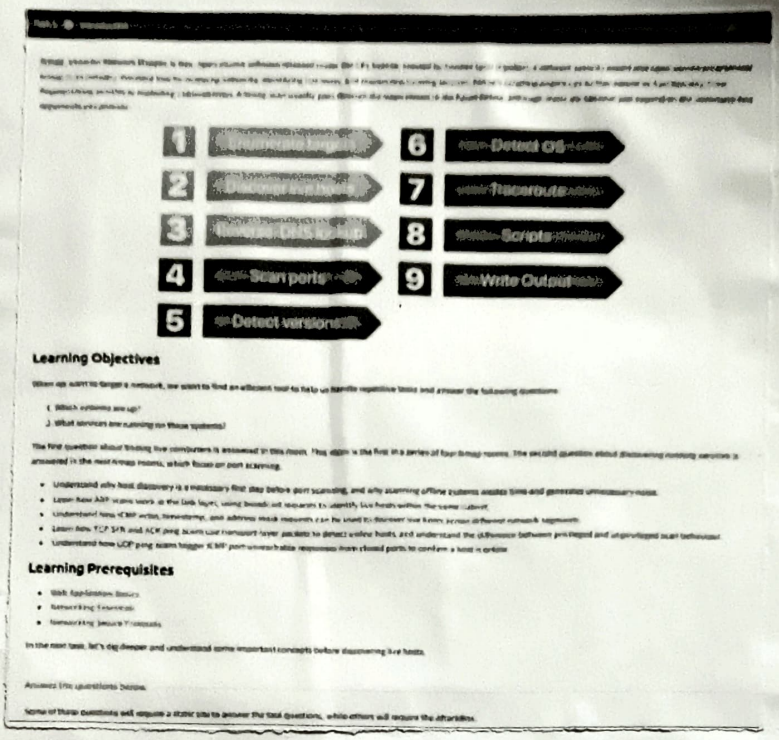
ICMP Address Mask (-PM):

```
$ nmap -Pm -Sn 10.200.6.0/24
Nmap done: 256 IP address (0 hosts up)
           Scanned in 52.14 seconds
```

TCP & UDP ping Discovery:

(i) TCP SVN Ping:

```
$ nmap -PS -Sn 10.200.6.0/24
Nmap Scan repost for 10.200.6.50
```



Learning Objectives

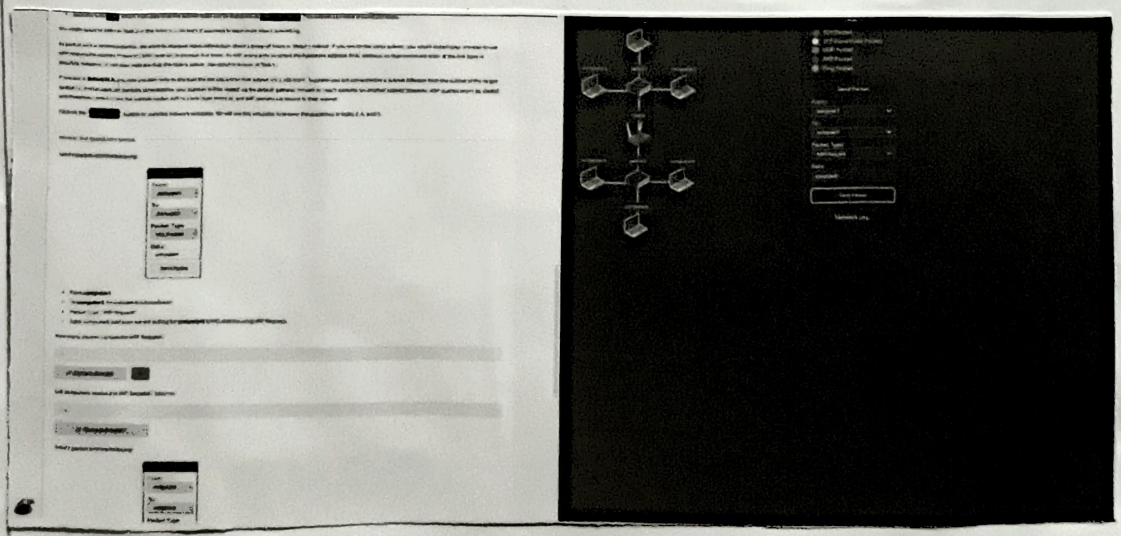
1. Which systems are up?
2. What services are running on those systems?

- Understand why host discovery is a necessary first step before port scanning, and why scanning offline systems wastes time and generates unnecessary noise.
- Learn how ARP scans work at the link layer, using broadcast requests to identify live hosts within the same subnet.
- Understand how ICMP echo, timestamp, and address mask requests can be used to discover live hosts across different network segments.
- Learn how TCP SYN and ACK ping scans use transport layer packets to detect online hosts, and understand the difference between privileged and unprivileged scan behaviour.
- Understand how UDP ping scans trigger ICMP port unreachable responses from closed ports to confirm a host is online.

Learning Prerequisites

In the next task, let's dig deeper and understand some important concepts before discovering live hosts.

Answer the questions below.



Host is up (0.070s latency)
Nmap scan repost for 10.200.6.250
Host is up (0.0 latency)

(ii) TCP Ack Ping (-PA)

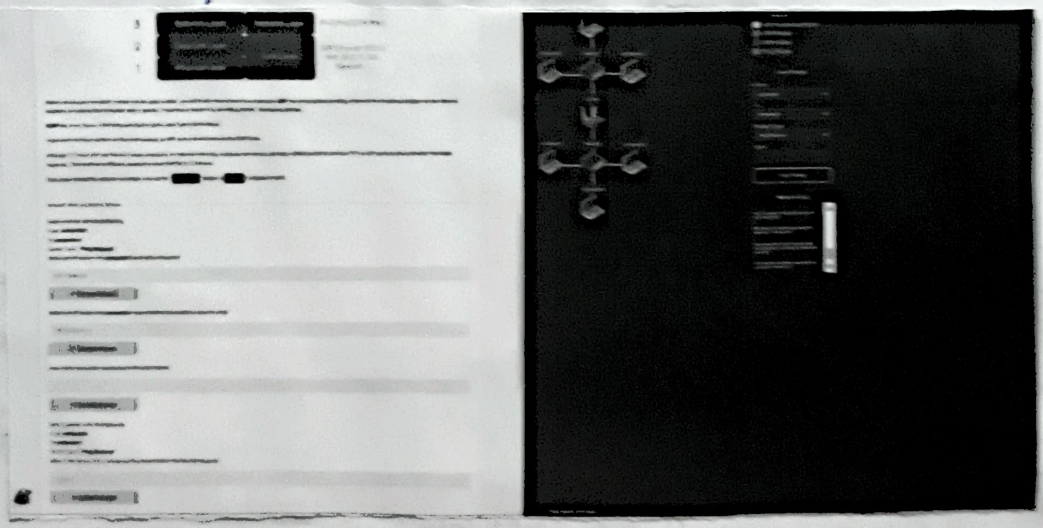
$ sudo nmap -PA -sn 10.200.6.0/24
Nmap scan repost for 10.200.6.50
Host is up (0.070s latency)
Nmap scan repost for 10.200.6.250

(iii) UDP Ping (-PU)

$ sudo nmap -PU -sn 10.200.6.0/24
Nmap scan repost for 10.200.6.50
Host is up (0.070s latency)
Nmap scan repost for 10.200.6.250

Mass can -A faster Alternative:

$ masscan 10.200.6.0/24 -p 443
$ masscan 10.200.6.0/24 -ps on 443
$ masscan 10.200.6.0/24 -p22 -25



RESULT:

The live hosts on a network were discovered successfully using Nmap in Try hackme environment